Cryptography sheet # 2

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In this sheet, we continue playing with encryption techniques and the modular arithmetic and, also, try to get a feel of how to argue in abstract algebra.

1 Caesar strikes back

Caesar feels quite angry about the fact that his cleverly encrypted message has been intercepted and his cipher broken by Asterix and his friends. He decides to do something about that. Next time, he is going to use another encryption. One of Caesar's councelors, the wizard Xi from China, told him that Chinese scholars have been thoroughly studying modular arithmetic in the last 1000 years and that there is more power to it than the ability to add, as used in Caesar's cipher.

Xi also secretly thinks to himself that those Romans, considered so wise, could not come up with anything more advanced than the absolutely elementary function $\mathrm{f}(x) = x + b$ over $\mathbb{Z}_{26}$ to reshuffle the alphabet. As if one would have only learned to add in school. There is also multiplication. "And everyone considers Romans so cool – they are just arrogant boasters!"

So, Xi suggests to apply a more intricate function $\mathrm{f}(x) = ax + b$ over $\mathbb{Z}_{26}$ on the symbols of the plain text to get the cipher text. One would add b as before, but now there is a factor a in front of x, which introduces additional diversity. Caesar is exited. He also vaguely remembers seeing functions like $ax + b$ when he was a school kid, but in those days, they were not over finite rings. What a luck to have a Chinese councelor around! There is a technical problem though. The function f should be chosen the way that the alphabet is really reshuffled and it would also be possible to decipher by applying the inverse function. In technical terms this means that $\mathrm{f}\colon \mathbb{Z}_{26} \to \mathbb{Z}_{26}$ should be bijective: different inputs are mapped to different outputs, and every element in the range is generated as the output for some choice of the input. Describe all choices of $a, b \in \mathbb{Z}_{26}$ would give a bijective function. Remark: you'll quickly see that b does not make any problems, but there are different choices of a - some will work, some not.

2 Asterix never gives up

Imagine you are Asterix, and you intercept another encrypted message from Caesar. This time the message is "uqpxfubsuqpjfu". You think that, once again, this message might be encrypted with Caesar's favorite cipher, but after applying the brute force attack under this assumption you do not see any reasonable deciphered messages. It must be another type of cipher then... What else could one do? You ask Getafix for advice and he suggests that Ceasar finally learned multiplication and might have used the function $\mathrm{f}(x) = ax + b$ over $\mathbb{Z}_{26}$ to reshuffle the alphabet. For this function, too, one could carry out a brute force attack, but this would be much more laborious: one needs to guess both a and b. Going over all possible pairs of (a, b) would give $26^2 = 676$ possibilities. If you exclude the choices of a that are not reasonable, you would reduce that number, but it still will be quite large. You are impatient and do not want to spend so much time on checking all these possibilities. Luckily, Getafix gives you another piece of advice. Caesar has written his message in English again. And, in contrast to Latin, English has articles like "the". Consequently, in English messages, one is likely to see triples of symbols repeated many times. If you detect such a triple and assume it encrypts "the", could you then determine a and b and get the message? Try doing this.

3 Unique negative

Having seen cryptographic applications, you start to like the modular ring Z_n more and more, but you also start appreciating abstract rings $(R, +, \cdot)$, too. If you verify something about abstract rings, you verify it it about every possible ring! and, who knows, maybe there are other rings apart from $\mathbb{Z}_n$ that may be useful. One of the properties that defined unitary commutative rings $(R, +, \cdot)$ was that to each $a \in R$ there exists an element $b \in R$ satisfying $a + b = 0$. It turns out that this element is always unique, for a given a. This means, that if you have $b \in R$ and some $c \in R$, both related to a in the same manner $a + b = a + c = 0$, then one necessarily has $b = c$. Can you explain, why this is true? Remember that $+$ is just an abstract operation, for which some properties are assumed to be true. Since the exercise is kind of basic, it makes sense to be very explicit regarding what property you use in each of the steps of your derivation. If you have a feeling that $a + b = a + c = 0$ implies $b = c$ is obvious, think again that $+$ is an abstract operation, and that the uniqueness of the negative element would not be true for any operation (but it is true for the $+$ of our abstract ring). If your operation would be the binary maximum on $\mathbb{Z}$, the property wouldn't be valid: $\max\{a, b\} = \max\{a, c\} = 0$ does not always imply $b = c$, since one can have $a = 0, b = -1$ and $c = -2$. Your response could be: but maximum is not a plus. The response of abstract algebraists to such a message is this: Why not? In principle, when you create an algebraic structure and fix operations, your operation can be anything and you can give it a name and notation as you like, calling it $+, \cdot, *, \circ, \#$ etc. On the other hand, it is clear, that max would not qualify as a plus of a unitary commutative ring, because it does not satisfy some of the properties required by the definition.

4 A few insights about the groups

There is a much "slimmer" algebraic structure than the ring. This structure is group. Groups are defined through just one binary operation (in contrast to rings, which need two). A group is a structure $(G, *)$ with a binary operation $*\colon G \times G \to G$ such that G has an element e satisfying $a * e = e * a = a$ for every $a \in G$ and apart from that, one has $(a * b) * c = a * (b * c)$ for every $a, b, c \in G$ and, for every $a \in G$, there exist a $b \in G$ satisfying $a * b = e$. The group is called Abelian if $a * b = b * a$ for all $a, b \in G$. The choice of the notation for the group operation is flexible: some use $\circ$, some others $\cdot$, and when the group is Abelian, it is very common to use $+$. The element e is called the neutral element of the group. When the group operation is denoted by $\cdot$, one uses the notation $e = 1$, and when it is doted by $+$, one uses the notation $e = 0$.

1. Show that the neutral element is unique,
2. show that for every $a \in G$, the element b satisfying $a * b = e$ is unique and also satisfies the equation $b * a =$ e. This unique b is denoted as a^{-1} , but when the group operation is denoted by $+$, one uses the notation $-a$.